

A Appendix

A.1 Proof of Theorem 5.1

For notational convenience, we will denote $f_{q,d}(D') := \max_{d' \in q(D')} \text{sim}(d, d')$.

PROOF.

1. Normalized non-negativity: This is an immediate consequence of the fact that $\text{sim}(d, d') \in [0, 1] \forall d, d' \in D$.

2. Monotonicity: $A \subseteq B \subseteq D \implies q(A) \subseteq q(B) \subseteq q(D) \implies f_{q,d}(A) = \max_{d' \in q(A)} \text{sim}(d, d') \leq \max_{d' \in q(B)} \text{sim}(d, d') = f_{q,d}(B) \implies f(A) \leq f(B)$. The last implication follows from the fact that sums and expectations respect monotonicity.

3. Submodularity: Given any $A \subseteq B \subseteq D$ and $d^* \in D \setminus B$, we distinguish two cases:

• **Case 1:** $q(\{d^*\}) = \emptyset$. In this scenario,

$$\begin{aligned} \Delta_{f_{q,d}}(d^*|B) &= \max_{d' \in q(B \cup \{d^*\})} \text{sim}(d, d') - \max_{d' \in q(B)} \text{sim}(d, d') \\ &= \max_{q(B \cup \{d^*\}) = q(B) \cup q(\{d^*\}) = q(B)} 0 \\ &= \Delta_{f_{q,d}}(d^*|A). \end{aligned}$$

• **Case 2:** $q(\{d^*\}) = d^*$. In this scenario,

$$\begin{aligned} \Delta_{f_{q,d}}(d^*|B) &= \max_{d' \in q(B \cup \{d^*\})} \text{sim}(d, d') - \max_{d' \in q(B)} \text{sim}(d, d') \\ &= \max\left\{0, \text{sim}(d, d^*) - \max_{d' \in q(B)} \text{sim}(d, d')\right\}. \end{aligned}$$

Now, for any $d \in D$,

$$\begin{aligned} A \subseteq B &\implies q(A) \subseteq q(B) \\ &\implies \max_{d' \in q(A)} \text{sim}(d, d') \leq \max_{d' \in q(B)} \text{sim}(d, d') \\ &\implies -\max_{d' \in q(B)} \text{sim}(d, d') \leq -\max_{d' \in q(A)} \text{sim}(d, d') \\ &\implies \text{sim}(d, d^*) - \max_{d' \in q(B)} \text{sim}(d, d') \\ &\leq \text{sim}(d, d^*) - \max_{d' \in q(A)} \text{sim}(d, d'). \end{aligned}$$

Hence, $\Delta_{f_{q,d}}(d^*|B) \leq \Delta_{f_{q,d}}(d^*|A)$.

This proves that functions $f_{q,d}(D')$ are submodular. The submodularity of $f(D')$ follows from the fact that both the sum and expectation operators preserve submodularity [22]. $\square$

A.2 Proof of Theorem 5.2

LEMMA A.1. For any given datasets $D' \subseteq D$ and monotone SPJU query q ,

$$|q(D')| = \sum_{d \in D'} |q(\{d\})|.$$

PROOF. $D' = \cup_{d \in D'} \{d\} \xRightarrow{(1.1)} q(D') = q(\cup_{d \in D'} \{d\}) \xRightarrow{(1.2)} q(D') = \cup_{d \in D'} q(\{d\}) \xRightarrow{(1.3)} |q(D')| = |\cup_{d \in D'} q(\{d\})| \xRightarrow{(1.4)} |q(D')| = \sum_{d \in D'} |q(\{d\})|$. ($\cup$ denotes the disjoint union). (1.1) holds because the answer set of any query q on any subset $D' \subseteq D$ is unique. (1.2) holds because q distributes under set union. (1.3) holds because equal sets have the same cardinality. (1.4) holds because of the inclusion-exclusion principle in the case of a disjoint union. $\square$

To prove that the objective is double-stochastic we must prove that there exists a probability distribution $\mathcal{D}$ such that $f(D') = \mathbb{E}_{(d,q) \sim \mathcal{D}} f_{q,d}(D')$.

$$\begin{aligned}
f(D') &= \mathbb{E}_{q \sim Q} [S(q(D), q(D'))] \\
&= \mathbb{E}_{q \sim Q} \left[\frac{1}{|q(D)|} \cdot \sum_{d \in q(D)} \max_{d' \in q(D')} \text{sim}(d, d') \right] \\
&= \sum_{q \in Q} \mathbb{P}_Q(q) \cdot \frac{1}{|q(D)|} \sum_{d \in q(D)} f_{q,d}(D') \\
&= \sum_{q \in Q} \mathbb{P}_Q(q) \sum_{d \in q(D)} \frac{1}{|q(D)|} \cdot f_{q,d}(D') \quad (*) \\
&= \sum_{q \in Q} \mathbb{P}_Q(q) \left(\sum_{d \in q(D)} \frac{1}{|q(D)|} \cdot f_{q,d}(D') + \sum_{d \in D \setminus q(D)} 0 \right) \\
&= \sum_{q \in Q} \mathbb{P}_Q(q) \sum_{d \in D} \frac{\mathbb{I}_{\{d \in q(D)\}}(d)}{|q(D)|} \cdot f_{q,d}(D') \\
&= \sum_{q \in Q} \mathbb{P}_Q(q) \sum_{d \in D} \frac{|q(\{d\})|}{|q(D)|} \cdot f_{q,d}(D') \\
&= \sum_{q \in Q} \sum_{d \in D} \mathbb{P}_Q(q) \cdot \frac{|q(\{d\})|}{|q(D)|} \cdot f_{q,d}(D') \\
&= \sum_{q \in Q} \sum_{d \in D} \mathbb{P}_{\mathcal{D}}(d, q) \cdot f_{q,d}(D') \\
&= \sum_{d \in D} \sum_{q \in Q} \mathbb{P}_{\mathcal{D}}(d, q) \cdot f_{q,d}(D') \\
&= \mathbb{E}_{(d,q) \sim \mathcal{D}} [f_{q,d}(D')].
\end{aligned}$$

Hence, $f(D') = \mathbb{E}_{(d,q) \sim \mathcal{D}} f_{q,d}(D')$ for $\mathcal{D}$ with probability mass function $\mathbb{P}_{\mathcal{D}}(d, q) = \frac{|q(\{d\})|}{|q(D)|} \cdot \mathbb{P}_Q(q)$, $\forall (d, q) \in D \times Q$. Last, we show this is indeed a probability mass function:

1. Normalized non-negativity: $\mathbb{P}_{\mathcal{D}}(d, q) \geq 0$: This is true since $\frac{|q(\{d\})|}{|q(D)|} \geq 0$ and $\mathbb{P}_Q(q) \in [0, 1]$. Note that queries are assumed to be non-empty (i.e., $q(D) \neq \emptyset$, so no division by 0 takes place).

2. Sum up to 1: $\sum_{(d,q) \in D \times Q} \mathbb{P}_{\mathcal{D}}(d, q) = 1$:

$$\begin{aligned}
\sum_{(d,q) \in D \times Q} \mathbb{P}_{\mathcal{D}}(d, q) &= \sum_{d \in D} \sum_{q \in Q} \mathbb{P}_{\mathcal{D}}(d, q) = \sum_{q \in Q} \sum_{d \in D} \mathbb{P}_{\mathcal{D}}(d, q) \\
&= \sum_{q \in Q} \sum_{d \in D} \frac{|q(\{d\})|}{|q(D)|} \cdot \mathbb{P}_Q(q) = \sum_{q \in Q} \mathbb{P}_Q(q) \sum_{d \in D} \frac{|q(\{d\})|}{|q(D)|} \\
&= \sum_{q \in Q} \mathbb{P}_Q(q) \frac{\sum_{d \in D} |q(\{d\})|}{|q(D)|} \stackrel{\text{Lemma A.1}}{=} \sum_{q \in Q} \mathbb{P}_Q(q) \frac{|q(D)|}{|q(D)|} = 1.
\end{aligned}$$

Next, we prove the second part of the theorem, i.e., the fact that estimating $\nabla F_{q,d}(\mathbf{x})$ is less costly than estimating $\nabla F_q(\mathbf{x})$. Recall that $F_{q,d}(\mathbf{x})$ denotes the multilinear extension of $f_{q,d}(D') = \max_{d' \in q(D')} \text{sim}(d, d')$, and $F_q(\mathbf{x})$ corresponds to the multilinear extension of $f_q(D') = S(q(D), q(D'))$ with $S(\cdot, \cdot) = (2)$. The equation marked as (*) implies that

$$f(D') = \mathbb{E}_{q \sim Q} \mathbb{E}_{d \sim \text{unif}(q(D))} [f_{q,d}(D')].$$

Therefore, $f_q(D') = \mathbb{E}_{d \sim \text{unif}(q(D))} [f_{q,d}(D')]$. Further, by the linearity of the multilinear extension [41], $F_q(\mathbf{x}) = \mathbb{E}_{d \sim \text{unif}(q(D))} [F_{q,d}(\mathbf{x})]$. Since ∇ is linear, $\nabla F_q(\mathbf{x}) = \mathbb{E}_{d \sim \text{unif}(q(D))} [\nabla F_{q,d}(\mathbf{x})]$. Given that

queries are assumed to be non-empty, $|q(D)| \geq 1$. Thus, estimating $\nabla F_q(\mathbf{x})$ requires estimating at least as many terms as estimating $\nabla F_{q,d}(\mathbf{x})$.

A.3 Proof of Theorem 5.6

If $S(q(D), q(D')) = \text{Jaccard}(q(D), q(D')) = \text{Precision}(q, D', D)$, objective 1 becomes finding a subset $D^* \subseteq D$ such that

$$\begin{aligned}
D^* &= \arg \max_{D' \subseteq D, |D'| \leq B} \mathbb{E}_{q \sim Q} \left[\frac{|q(D')|}{|q(D)|} \right] \\
&= \arg \max_{D' \subseteq D, |D'| \leq B} \sum_{q \in Q} \mathbb{P}_Q(q) \cdot \frac{|q(D')|}{|q(D)|} \\
&\stackrel{\text{Lemma A.1}}{=} \arg \max_{D' \subseteq D, |D'| \leq B} \sum_{q \in Q} \mathbb{P}_Q(q) \cdot \frac{\sum_{d \in D'} |q(\{d\})|}{|q(D)|} \\
&= \arg \max_{D' \subseteq D, |D'| \leq B} \sum_{q \in Q} \mathbb{P}_Q(q) \cdot \sum_{d \in D'} \frac{|q(\{d\})|}{|q(D)|} \\
&= \arg \max_{D' \subseteq D, |D'| \leq B} \sum_{q \in Q} \sum_{d \in D'} \mathbb{P}_Q(q) \cdot \frac{|q(\{d\})|}{|q(D)|} \\
&= \arg \max_{D' \subseteq D, |D'| \leq B} \sum_{d \in D'} \sum_{q \in Q} \mathbb{P}_Q(q) \cdot \frac{|q(\{d\})|}{|q(D)|} \\
&= \arg \max_{D' \subseteq D, \sum_{d \in D'} 1 \leq B} \sum_{d \in D'} \mathbb{E}_{q \sim Q} \left[\frac{|q(\{d\})|}{|q(D)|} \right].
\end{aligned}$$

Hence, the optimal solution to the independent data forgetting problem corresponds to those B elements in D that have the highest score according to $\text{score}(d) = \mathbb{E}_{q \sim Q} \left[\frac{|q(\{d\})|}{|q(D)|} \right]$.

This result extends beyond the simple cardinality constraint. In particular, it holds for the (more general) Knapsack constraint. If the cost of each tuple $d \in D$ is given by a cost function $C : D \rightarrow \mathbb{R}_+$ where $C(D') := \sum_{d \in D'} C(d) \forall D' \subseteq D$, the problem reduces to the 0-1 Knapsack problem where the set of n items corresponds to the dataset D , $v_d = \mathbb{E}_{q \sim Q} \left[\frac{|q(\{d\})|}{|q(D)|} \right]$, $w_d = C(d)$, and $W = B$.

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